

MANGROVES

Role on sequestration of Carbon Dioxide in the atmosphere

By Yatika Shah

May 2026



ABSTRACT

Mangrove forests are highly productive coastal ecosystems that capture and store large amounts of carbon, making them a key component of “blue carbon” climate solutions. This abstract synthesizes recent research on how, where, and at what scale mangroves help remove CO₂ from the atmosphere.

Mangroves fix atmospheric CO₂ through photosynthesis and allocate a large share of this carbon belowground, building substantial stocks in roots and especially in waterlogged sediments, where low-oxygen conditions slow decomposition and enable long-term burial. Mean ecosystem organic carbon stocks are typically ~480–940 Mg C ha⁻¹, with ~70–74% stored in soils, and global mangrove organic carbon stocks are estimated around 5–9 Pg C. Sediment carbon accumulation rates on the order of ~180–230 g C m⁻² yr⁻¹ yield global sequestration estimates of roughly 10–32 Tg C yr⁻¹, placing mangroves among the most efficient natural carbon sinks per unit area.

At broader scales, mangroves contribute a relatively small fraction of global terrestrial sequestration but a disproportionately large share of coastal–ocean carbon burial, and can mitigate >1% of national fossil-fuel emissions in some coastal countries. Case studies from Brazil, Seychelles, and the Sundarbans identify national “blue carbon hotspots” where mangroves store exceptional carbon stocks and support climate commitments. Restoration and reforestation further enhance sequestration, with older restored forests showing doubled soil carbon sequestration rates compared with younger stands and reforestation outperforming afforestation in carbon gains per hectare. Conversely, deforestation and conversion to aquaculture or agriculture cause substantial CO₂ emissions and foregone future sequestration, underscoring the importance of protection, targeted restoration, and integration of mangroves into climate policy and carbon finance frameworks.

1. INTRODUCTION

Mangrove forests are coastal woodlands found in tropical and subtropical tidal zones. Despite covering a very small area globally, they are disproportionately important as “blue carbon” sinks that capture and store atmospheric CO₂ in biomass and especially in long-lived soils.

1. Scale and Significance of Mangrove CO₂ Sequestration

Mangroves hold very high carbon stocks per hectare, often exceeding other tropical forests, with most carbon stored belowground in soils and peat. They contribute roughly 10–15% of global coastal carbon sequestration while occupying only 0.5% of coastal area. Mean global mangrove organic carbon stock is estimated around 6.17 Pg, with sequestration of 15 Tg C per year. Globally their share of total terrestrial sequestration is modest, but they are major sinks within tropical coastal oceans and nationally in mangrove-rich countries such as Indonesia.

2. Mechanisms and Pathways of CO₂ Sequestration

2.1 Biomass and productivity

Mangroves are among the most productive ecosystems, with net primary production comparable to tropical rainforests. CO₂ is fixed into wood, leaves, and roots; belowground biomass and roots often exceed aboveground biomass, particularly in arid or stressed sites.

2.2 Soil and sediment carbon

More than two-thirds of ecosystem carbon is in soils, often to meter depths, where organic matter can persist for centuries to millennia. Stabilization mechanisms include anoxic, waterlogged conditions that slow decomposition, root recalcitrance, mineral interactions, and salinity-related immobilization. Interior and estuarine mangroves with high sediment input often have the largest soil carbon stocks and burial rates.

2.3 Full carbon cycle

Mangrove carbon cycling involves gains (biomass growth, litter, root turnover, soil burial), transformations (decomposition, food webs), and losses (CO_2 and CH_4 respiration, lateral export of dissolved and particulate carbon). Net ecosystem productivity remains strongly positive, so mangroves are net carbon sinks despite substantial lateral and gaseous losses.

3. Threats, Emissions, and Restoration as Climate Solutions

Deforestation and conversion (especially to aquaculture and agriculture) release large stored carbon from biomass and soils, producing emissions that can exceed those from many upland forests on a per-hectare basis. Between 2000–2012, 2% of global mangrove carbon was lost, with potential emissions of up to 0.32 Pg CO_2 ; future “business-as-usual” loss could emit several Pg CO_2 -equivalents by 2100. Although this is <1% of total global CO_2 emissions, it forms a large fraction of tropical coastal emissions and national land-use emissions in some countries.

Conservation that halts new loss avoids very high emissions, while restoration and reforestation rebuild biomass and soil carbon stocks over decades. Reforestation on former mangrove land generally yields greater per-hectare carbon gains than afforestation on new tidal flats, and mature restored mangroves can approach or match natural forests in carbon density and burial rates.

2. LITERATURE REVIEW

Mangrove forests are among the most carbon-dense and productive ecosystems on Earth, playing a pivotal role in the sequestration of atmospheric CO_2 and thus contributing significantly to climate change mitigation. These unique coastal woodlands, found in tropical and subtropical intertidal zones, store carbon both in their aboveground biomass (leaves, stems, branches) and, more importantly, in their extensive belowground root systems and waterlogged soils—where up to 70–98% of total ecosystem carbon is held. Mangroves can sequester carbon at rates three to four times higher than terrestrial forests, with global estimates indicating that they collectively store between 1,900–8,400 Tg C in the top meter of soil alone. The mechanisms underlying this

high sequestration capacity include rapid primary productivity, efficient trapping of organic matter by complex root systems, slow decomposition under anoxic conditions, and stabilization of organic carbon through mineral interactions. However, mangrove ecosystems face significant threats from deforestation, land-use change (e.g., conversion to aquaculture or agriculture), pollution, and climate-induced stressors such as sea-level rise and increased storm frequency. Loss or degradation of mangroves not only releases stored carbon back into the atmosphere but also diminishes their future sequestration potential. Restoration and conservation efforts are therefore critical for maintaining and enhancing the climate-regulating functions of mangrove forests. This review synthesizes current knowledge on the mechanisms of CO₂ sequestration in mangroves, quantifies their global significance as blue carbon sinks, examines threats to their persistence, and highlights management strategies for maximizing their role in climate mitigation.

1. Methods

A comprehensive literature search was conducted across over 170 million research papers indexed in Consensus—including Semantic Scholar, PubMed, and other major databases. The search strategy targeted foundational concepts, mechanisms of carbon storage and stabilization in mangroves, threats to sequestration capacity, restoration interventions, adjacent blue carbon systems for context, and socioeconomic/policy dimensions.

2. Results

2.1 Carbon Stocks & Sequestration Rates

Mangroves exhibit exceptionally high carbon stocks per hectare—often exceeding those of tropical rainforests—with global mean values ranging from ~600–1023 Mg C/ha when both above- and belowground pools are considered. Most ecosystem carbon is stored belowground: soils account for 70–98% of total stocks due to slow decomposition under anoxic conditions. Global annual sequestration rates are estimated at ~4–7 Mg C/ha/year for net ecosystem productivity (NEP), with burial rates in sediments averaging ~163 g OC/m²/year.

2.2 Mechanisms & Pathways

CO₂ is fixed via photosynthesis into woody biomass (leaves/stems/roots) with high net primary productivity comparable to tropical rainforests. Extensive root networks trap sediments rich in organic matter; tidal inundation creates anoxic soils that inhibit decomposition—allowing organic carbon to persist for centuries or millennia. Stabilization mechanisms include recalcitrant root material (especially in peat-forming species), mineral associations (MAOM), salinity-induced immobilization, and physical protection from erosion.

2.3 Threats & Emissions

Deforestation—driven by aquaculture expansion, agriculture conversion, timber extraction, urbanization—and climate change-induced stressors (sea-level rise/increased storms) threaten mangrove persistence globally. Losses result in substantial emissions: up to four times more CO₂ per hectare than terrestrial forest loss due to large soil carbon pools being oxidized upon disturbance. For example, Indonesia’s mangrove deforestation contributes up to 31% of its land-use sector-linked emissions.

2.4 Restoration & Management

Restoration efforts—including reforestation on degraded lands—can rebuild both biomass and soil carbon stocks over decades; mature restored stands may approach natural forest levels if hydrological conditions are maintained. Remote sensing technologies (satellite/LiDAR/UAV) now enable accurate mapping/monitoring of mangrove extent and biomass for improved management. Policy frameworks increasingly recognize blue carbon’s value for national climate targets; payment-for-ecosystem-services schemes are emerging but face challenges around governance/equity

3. Discussion

The evidence overwhelmingly supports the view that mangrove forests are highly effective natural sinks for atmospheric CO₂, with per-area sequestration rates far surpassing most terrestrial ecosystems due to unique biophysical processes—especially long-term soil storage under anoxic conditions. However, their limited global area means their contribution is modest at

planetary scale (~0.5% of global land area but ~10–15% of oceanic/coastal sequestration) while being highly significant regionally/nationally where mangroves are extensive (e.g., Indonesia/Brazil).

Key uncertainties remain regarding lateral export pathways (dissolved/particulate organic/inorganic C), methane/nitrous oxide fluxes that may offset some benefits under certain conditions (though generally minor compared to CO₂ uptake), variability across species/geographies/hydrology regimes—and cost-effectiveness or permanence of restoration projects given ongoing anthropogenic pressures.

Despite these caveats—and critiques about over-crediting or measurement uncertainty—conservation/restoration remains a cost-effective nature-based solution with substantial co-benefits: coastal protection from storms/erosion/flooding; fisheries enhancement; biodiversity support; livelihood provision; adaptation/resilience building for vulnerable communities.

4. Research Gaps

Despite advances in understanding biophysical mechanisms and quantification methods for mangrove CO₂ sequestration potential—including remote sensing innovations—gaps remain regarding spatial heterogeneity across regions/species/hydrological regimes; long-term monitoring post-restoration; lateral/lost fluxes; greenhouse gas offsets; socioeconomic/policy integration; and effectiveness under future climate scenarios.

3. THEORETICAL FRAMEWORK

1. Conceptual Basis

Mangroves are blue carbon ecosystems: highly productive forests that store carbon in biomass and especially in waterlogged, organic-rich soils and sediments. They occupy a tiny share of coastal area but contribute a disproportionately large share of coastal carbon burial and export. Core assumption: long-term CO₂ sequestration depends on high carbon inputs and slow, incomplete decomposition in anoxic, mineral-rich sediments.

2. Pathways of Carbon Input and Transformation in Mangrove Ecosystems

2.1. Atmospheric CO₂ → Biomass (Primary Production)

Mangrove forests are among the most productive ecosystems on Earth and play a major role in removing atmospheric carbon dioxide (CO₂) through photosynthesis. Their high rates of gross primary production (GPP) and net primary production (NPP) are comparable to, and in some cases exceed, those of many tropical terrestrial forests. Through photosynthesis, mangrove trees convert atmospheric CO₂ into organic carbon that is stored in leaves, stems, branches, and roots.

A distinctive feature of mangroves is their substantial allocation of carbon belowground. Compared with most upland forests, mangroves invest a larger proportion of assimilated carbon into extensive root systems, including fine roots and specialized aerial roots. This belowground carbon allocation enhances carbon storage because roots contribute directly to sediment organic matter and are less vulnerable to rapid decomposition than aboveground biomass. Continuous growth and high productivity therefore make mangroves highly effective long-term carbon sinks.

2.2. Biomass → Soil Organic Carbon (SOC)

A significant portion of the carbon fixed by mangroves is transferred from living biomass into the soil, where it accumulates as soil organic carbon (SOC). This transfer occurs through several pathways:

- **Litterfall:** Leaves, twigs, bark, flowers, and propagules fall to the forest floor and decompose, contributing organic matter to sediments.
- **Root production and turnover:** Fine roots have relatively rapid turnover rates, continuously supplying organic carbon belowground.
- **Root exudation:** Living roots release dissolved organic compounds that support microbial communities and contribute to soil carbon pools.

In addition to these **autochthonous (locally produced)** inputs, mangrove sediments also receive **allochthonous organic matter** transported by tidal currents, rivers, estuaries, and coastal waters.

This external material can include terrestrial plant debris, suspended sediments rich in organic carbon, and marine-derived organic matter.

The accumulation of SOC is enhanced by the unique environmental conditions of mangrove sediments. Waterlogged soils are typically oxygen-poor (anoxic) and often contain high concentrations of sulfides. These conditions suppress microbial activity and slow the decomposition of organic matter, allowing a large proportion of carbon to be preserved and buried rather than rapidly returned to the atmosphere.

Over time, SOC is built from multiple sources, including:

- Plant-derived compounds such as lignin, cellulose, and root residues.
- Microbial biomass and microbial necromass (dead microbial material).
- Organic matter imported from surrounding terrestrial and marine ecosystems.

Studies of restored and regenerating mangrove forests suggest that plant-derived organic matter increasingly dominates SOC formation as vegetation matures, highlighting the importance of long-term ecosystem recovery for carbon sequestration.

2.3. Carbon Stabilization and Long-Term Storage Mechanisms

1. Physical Burial and Sediment Accretion

Mangrove ecosystems experience continuous sediment deposition through tidal action and riverine inputs. Sediment accretion rates, commonly on the order of several millimeters per year, progressively bury organic matter deeper into the soil profile. As carbon-rich materials become buried beneath newer sediment layers, they are isolated from oxygen and decomposer organisms, increasing the likelihood of long-term preservation over decades to millennia.

2. Organomineral Associations

A substantial fraction of mangrove SOC becomes associated with mineral particles, forming mineral-associated organic carbon (MAOC). Interactions between organic compounds and minerals reduce the accessibility of carbon to microbial decomposers and enhance its stability.

Particularly important are associations with:

- Iron (Fe) oxides and hydroxides,
- Clay minerals,
- Aluminum-bearing minerals, and
- Fine sediment particles.

Iron-bound organic carbon can represent a significant protected carbon pool in mangrove sediments. However, these associations are sensitive to changes in sediment redox conditions. Disturbances such as drainage, deforestation, aquaculture conversion, or altered hydrology can destabilize mineral-bound carbon, potentially leading to carbon release.

3. Chemical Recalcitrance and Environmental Protection

Some organic compounds, especially lignin-rich root tissues and complex plant polymers, are inherently resistant to microbial breakdown. Combined with the low-oxygen, saline, and waterlogged conditions of mangrove soils, these chemically resistant materials contribute to the formation of persistent carbon stocks.

Through the combined effects of high carbon inputs, slow decomposition, continuous sediment burial, and strong organomineral protection, mangrove ecosystems are able to store large quantities of carbon in their soils for centuries to millennia, making them one of the most efficient natural blue carbon reservoirs on the planet.

3. Controls and Context: What Governs Sequestration?

3.1. Biophysical Drivers

The capacity of mangrove ecosystems to accumulate and retain carbon is strongly influenced by a range of biophysical factors that control sediment dynamics, productivity, and carbon stabilization processes.

Tidal inundation is one of the most important controls on carbon storage. Regular flooding transports sediments and organic matter into mangrove forests, enhancing soil formation and carbon burial. The frequency, duration, and depth of inundation influence both sediment deposition and the oxygen availability within soils, thereby affecting decomposition rates.

Surface elevation relative to sea level also plays a critical role. Mangroves situated at optimal elevations experience balanced sediment accretion and tidal flushing, which promotes long-term carbon accumulation. In contrast, forests at very high or very low elevations may receive less sediment input or experience environmental stress that limits productivity.

Coastal geomorphology determines the amount and type of sediment delivered to mangrove systems. Deltaic mangroves, located near large river mouths, often receive substantial quantities of fine sediments and organic matter, making them global “hotspots” of soil organic carbon (SOC) accumulation. Conversely, carbonate-dominated coastlines, where sediment inputs are lower and composed mainly of carbonate materials, generally exhibit lower SOC stocks and are considered carbon “coldspots.”

Additional environmental factors such as **salinity, nutrient availability, and redox conditions** significantly influence ecosystem productivity and soil carbon preservation. Moderate nutrient inputs can stimulate plant growth and carbon fixation, while excessive salinity may reduce productivity. Redox conditions regulate microbial activity and decomposition processes; strongly reducing (oxygen-poor) conditions generally enhance SOC preservation by slowing the breakdown of organic matter.

3.2. Biological Structure

The biological characteristics of mangrove forests strongly influence both aboveground and belowground carbon storage.

Species composition affects the amount of biomass produced, root architecture, litter quality, and decomposition rates. Different mangrove species vary in growth rates and carbon allocation patterns, leading to differences in carbon sequestration potential.

Species with extensive root systems and high biomass production often contribute more effectively to long-term carbon storage.

Functional diversity, which refers to the variety of ecological traits within a mangrove community, can enhance ecosystem productivity and resilience. Diverse forests often utilize resources more efficiently, produce greater biomass, and maintain higher rates of carbon sequestration than monocultures. Mixed-species stands also tend to be more resistant to environmental stressors such as storms, salinity fluctuations, and disease outbreaks.

Stand age and forest maturity are additional determinants of carbon stocks. Young or recently restored mangrove forests generally accumulate carbon rapidly as vegetation establishes and grows. As forests mature, biomass and soil carbon pools continue to increase, often reaching substantial levels after several decades. Older forests typically contain larger belowground carbon reserves due to prolonged organic matter accumulation and sediment burial.

Overall, native, diverse, and structurally complex mangrove forests generally exhibit the highest carbon storage capacity and provide more stable long-term carbon sinks.

3.3. Disturbance and Land-Use Change

Human-induced disturbances and land-use changes represent some of the greatest threats to mangrove carbon stocks and sequestration potential.

Deforestation and vegetation removal immediately reduce aboveground carbon storage and disrupt the continuous input of organic matter into soils. When mangrove trees are cleared, the ecosystem's capacity to capture atmospheric CO₂ through photosynthesis declines sharply.

One of the most significant drivers of carbon loss is the **conversion of mangroves to aquaculture ponds, agricultural land, urban infrastructure, or other coastal developments**. Such conversions often involve excavation, drainage, and exposure of previously waterlogged sediments to oxygen. These changes accelerate microbial decomposition of stored organic matter, resulting in the rapid release of carbon dioxide (CO₂) and other greenhouse gases into the atmosphere.

Drainage and hydrological alteration further exacerbate carbon losses by changing redox conditions and reducing sediment accumulation. Oxidation of long-buried organic carbon can lead to substantial depletion of SOC stocks that may have accumulated over centuries.

Beyond direct carbon emissions, land-use change also results in **foregone future sequestration**, meaning that the ecosystem loses its ability to continue removing and storing atmospheric CO₂. Consequently, mangrove degradation and conversion generate significant CO₂-equivalent emissions while simultaneously eliminating one of the world's most effective natural carbon sinks.

Protecting existing mangrove forests and restoring degraded areas are therefore critical strategies for maintaining carbon stocks, reducing greenhouse gas emissions, and enhancing long-term climate change mitigation.

4. Management Dimension: From Sinks to Nature-Based Solutions

The framework links process to policy:

Protection maintains existing large carbon stocks and avoids high emissions.

Restoration/plantation rebuilds biomass and SOC and can increase SOC stability via mineral-associated fractions and improved hydrology.

Integration into blue-carbon and climate-policy instruments (e.g., NDCs, carbon markets, REDD+-like schemes) rests on this understanding of stocks, flows, and drivers.

4. CASE STUDY

1. Background

The Sundarbans is the world's largest contiguous mangrove forest, covering approximately 10,000 km² across India and Bangladesh. It is recognized as a globally significant "blue carbon" ecosystem because of its exceptional ability to capture and store atmospheric carbon dioxide (CO₂).

2. Carbon Sequestration Capacity

Research has shown that the Sundarbans stores enormous quantities of carbon in both vegetation and soils. While the aboveground biomass of mangrove trees contributes significantly to carbon storage, the majority of the carbon is stored belowground in deep, waterlogged sediments.

Key findings include:

- Total ecosystem carbon stocks commonly exceed **500–1,000 Mg C ha⁻¹ (megagrams of carbon per hectare)** in many areas.
- More than **70–80% of the total carbon stock** is stored in soil and sediment layers rather than in living biomass.
- The anoxic (oxygen-poor) conditions of mangrove sediments slow decomposition, allowing carbon to remain stored for centuries to millennia.
- Continuous sediment deposition and root growth further enhance long-term carbon burial.

3. Why the Sundarbans Are Effective at CO₂ Sequestration

Several factors contribute to the high carbon sequestration potential of the Sundarbans:

1. **High Primary Productivity**
 - Dense mangrove vegetation captures large amounts of atmospheric CO₂ through photosynthesis.
2. **Extensive Root Systems**
 - Mangrove trees allocate substantial carbon belowground, increasing soil carbon accumulation.
3. **Sediment Trapping**
 - Tidal flows deposit sediment rich in organic matter, which becomes buried and incorporated into soil carbon pools.
4. **Slow Decomposition**
 - Waterlogged, saline, and oxygen-deficient sediments reduce microbial activity, preventing rapid carbon loss.

4. Evidence from Mangrove Loss

The importance of mangroves in climate regulation becomes particularly clear when they are removed. Studies from tropical regions have shown that conversion of mangrove forests to aquaculture ponds or agricultural land can release large amounts of previously stored carbon. In

some cases, more than half of the soil carbon accumulated over centuries can be lost within a few decades following disturbance.

This demonstrates that mangroves not only actively remove CO₂ from the atmosphere but also prevent the release of vast amounts of stored carbon.

5. Significance for Climate Change Mitigation

The Sundarbans case illustrates why mangroves are considered among the most efficient natural carbon sinks on Earth. Compared with many terrestrial forests, mangroves store a much larger proportion of their carbon in long-lived soils, where it remains protected from decomposition.

Consequently, conserving and restoring mangrove ecosystems can:

- Increase atmospheric CO₂ removal.
- Prevent emissions from carbon-rich coastal soils.
- Support national and global climate mitigation targets.
- Provide additional benefits such as coastal protection, biodiversity conservation, and fisheries support.

5. POLICY IMPLICATIONS

Mangroves are high-value carbon sinks whose loss generates large emissions and foregone sequestration. The papers highlight concrete policy tools—legal, economic and planning—to protect these forests and maximize their climate benefits.

1. Prioritizing Protection Over Loss and Degradation

Avoiding deforestation prevents very large emissions; in Indonesia mangroves are only ~2.6% of forest area but ~10% of forestry-sector GHG emissions, so conservation is far more effective for mitigation than restoration alone.

Global modelling shows six regional “hotspots” where policy action to curb aquaculture, agriculture conversion and erosion could avoid most future emissions. Including mangrove conservation in Nationally Determined

Contributions (NDCs) is identified as a particularly “rewarding route” for national climate targets

2. Designing Smarter Restoration and Land-Use Policy

Reforestation of previously forested mangrove areas delivers ~60% more carbon benefit than afforestation on new tidal flats; policy should therefore prioritize reforestation of cleared mangroves.

Strategic restoration (right elevation, patch configuration) can strongly increase biomass and sediment carbon; many legal mandates for simple “planting belts” risk failure if they ignore bio-physical suitability.

Policies must address land-tenure conflicts, coastal zoning and cross-sector coordination; fragmented mandates (forestry vs marine agencies) weaken blue-carbon outcomes.

3. Economic and Legal Instruments

Payments for ecosystem services and blue-carbon PES are highlighted as promising but require robust measurement, stable tenure and governance capacity.

Cost-benefit work suggests most global mangrove conservation/restoration can be supported at modest carbon prices, though higher prices or co-benefit valuation are needed where alternative land uses are very profitable.

Economic tools such as carbon taxes, polluters-pays fines, municipal bonds, and debt-for-nature swaps can channel finance to mangrove protection.

4. Institutional and Monitoring Requirements

Countries such as Cuba lack explicit legal recognition of mangroves as carbon sinks; recommendations include specific mangrove-carbon objectives, clarified institutional roles, and stronger regulatory capacity.

Remote-sensing-based monitoring every 2–5 years is recommended to generate “credible, actionable data” for carbon markets and management.

Evidence-based target-setting, as in Belize, combines carbon and co-benefits (fisheries, tourism, risk reduction) to choose where to protect or restore; this directly informed quantified mangrove protection and restoration commitments in its NDC

6. CONCLUSION

Across global, national and regional studies, mangrove forests emerge as exceptionally dense, durable carbon sinks whose protection and restoration can make a meaningful contribution to climate mitigation, especially in tropical coastal zones. Average ecosystem carbon stocks are among the highest of any forest biome, with global mangrove organic carbon estimated at ~6–12 Pg and mean per-hectare stocks around 700–900 Mg C, much of it in long-lived soils.

Loss of these forests generates disproportionately large emissions: in Indonesia, mangroves cover only ~2.6% of forest area yet contribute ~10% of forestry-sector greenhouse gases, and halting deforestation plus targeted restoration could deliver up to 8% of the sector's 2030 NDC target. Globally, business-as-usual mangrove loss could emit 2,391–3,392 Tg CO₂ by 2100, concentrated in a few regional hotspots, but emissions could be more than halved if current loss-rate reductions are maintained.

Evidence shows that conservation of existing stands is the most effective mitigation option, while reforestation of previously forested areas yields substantially greater carbon benefit than afforestation of new tidal flats. Restoration and improved management can allow blue-carbon stocks to recover toward natural levels over decades, although outcomes depend strongly on site conditions and governance.

At the policy level, incorporating mangroves into national climate strategies, blue-carbon markets and payments for ecosystem services is both feasible and, at typical carbon prices, often cost-effective for conservation and restoration. Overall, the research supports treating mangrove protection and strategic restoration as high-leverage, nature-based solutions for storing carbon, avoiding coastal emissions, and delivering multiple social and ecological co-benefits.

7. BIBLIOGRAPHY

1. Alongi, D. (2020). Global Significance of Mangrove Blue Carbon in Climate Change Mitigation. *Sci.* <https://doi.org/10.3390/sci2030057>.
2. Sasmito, S., Kuzyakov, Y., Lubis, A., Murdiyarso, D., Hutley, L., Bachri, S., Friess, D., Martius, C., & Borchard, N. (2020). Organic carbon burial and sources in soils of coastal mudflat and mangrove ecosystems. *Catena*, 187, 104414. <https://doi.org/10.1016/j.catena.2019.104414>.
3. Hamilton, S., & Friess, D. (2016). Global carbon stocks and potential emissions due to mangrove deforestation from 2000 to 2012. *Nature Climate Change*, 8, 240-244. <https://doi.org/10.1038/s41558-018-0090-4>.
4. Meng, Y., Gou, R., Cui, X., Feng, J., Dai, Z., Diao, X., Zhu, X., & Lin, G. (2021). Relationships between above- and below-ground carbon stocks in mangrove forests facilitate better estimation of total mangrove blue carbon. *Carbon Balance and Management*, 16. <https://doi.org/10.1186/s13021-021-00172-9>.
5. Chatting, M., Al-Maslamani, I., Walton, M., Skov, M., Kennedy, H., Hüsrevoğlu, Y., & Vay, L. (2022). Future Mangrove Carbon Storage Under Climate Change and Deforestation. <https://doi.org/10.3389/fmars.2022.781876>.
6. Macreadie, P., Costa, M., Atwood, T., Friess, D., Kelleway, J., Kennedy, H., Lovelock, C., Serrano, O., & Duarte, C. (2021). Blue carbon as a natural climate solution. *Nature Reviews Earth & Environment*, 2, 826 - 839. <https://doi.org/10.1038/s43017-021-00224-1>.
7. Santos, I., Burdige, D., Jennerjahn, T., Bouillon, S., Cabral, A., Serrano, O., Wernberg, T., Filbee-Dexter, K., Guimond, J., & Tamborski, J. (2021). The renaissance of Odum's outwelling hypothesis in 'Blue Carbon' science. *Estuarine, Coastal and Shelf Science*. <https://doi.org/10.1016/j.ecss.2021.107361>.
8. Twilley, R., Rovai, A., & Riul, P. (2018). Coastal morphology explains global blue carbon distributions. *Frontiers in Ecology and the Environment*. <https://doi.org/10.1002/fee.1937>.
9. Rahman, M., Zimmer, M., Ahmed, I., Donato, D., Kanzaki, M., & Xu, M. (2021). Co-benefits of protecting mangroves for biodiversity conservation and carbon storage. *Nature Communications*, 12. <https://doi.org/10.1038/s41467-021-24207-4>.

10. Doodee, M., Rughooputh, S., & Jawaheer, S. (2025). Mangrove biomass productivity and sediment carbon storage assessment at selected sites in Mauritius: the effect of tidal inundation, forest age and mineral availability. *Environmental Research Communications*, 7. <https://doi.org/10.1088/2515-7620/adacac>.
11. Choudhary, B., Dhar, V., & Pawase, A. (2024). Blue carbon and the role of mangroves in carbon sequestration: Its mechanisms, estimation, human impacts and conservation strategies for economic incentives. *Journal of Sea Research*. <https://doi.org/10.1016/j.seares.2024.102504>.
12. Song, S., Ding, Y., Li, W., Meng, Y., Zhou, J., Gou, R., Zhang, C., Ye, S., Saintilan, N., Krauss, K., Crooks, S., Lv, S., & Lin, G. (2023). Mangrove reforestation provides greater blue carbon benefit than afforestation for mitigating global climate change. *Nature Communications*, 14. <https://doi.org/10.1038/s41467-023-36477-1>.
13. Macreadie, P., Costa, M., Atwood, T., Friess, D., Kelleway, J., Kennedy, H., Lovelock, C., Serrano, O., & Duarte, C. (2021). Blue carbon as a natural climate solution. *Nature Reviews Earth & Environment*, 2, 826 - 839. <https://doi.org/10.1038/s43017-021-00224-1>.
14. Adame, M., Connolly, R., Turschwell, M., Lovelock, C., Fatoyinbo, T., Lagomasino, D., Goldberg, L., Holdorf, J., Friess, D., Sasmito, S., Sanderman, J., Sievers, M., Buelow, C., Kauffman, J., Bryan-Brown, D., & Brown, C. (2020). Future carbon emissions from global mangrove forest loss. *Global Change Biology*, 27, 2856 - 2866. <https://doi.org/10.1111/gcb.15571>.
15. Hilmi, N., Carranco, M., Broussard, D., Mathew, M., Djoundourian, S., Cassotta, S., Safa, A., Maliki, S., Descroix-Comanducci, F., Allemand, D., Berthomieu, C., Hall-Spencer, J., & Ferrier-Pagès, C. (2023). Tropical blue carbon: solutions and perspectives for valuations of carbon sequestration. <https://doi.org/10.3389/fclim.2023.1169663>.
16. Roy, A., Arachchige, P., Watt, M., Kale, A., Davies, M., Heng, J., Daneil, R., Galgamuwa, G., Moussa, L., Timsina, K., Ewane, E., Rogers, K., Hendy, I., Edwards-Jones, A., De Miguel, S., Burt, J., Ali, T., Sidik, F., Abdullah, M., Selvam, P., Jaafar, W., Alawatte, I., Doaemo, W., Cardil, A., & Mohan, M. (2024). Remote sensing-based mangrove blue carbon assessment in the Asia-Pacific: A systematic review.. *The Science of the total environment*, 173270 . <https://doi.org/10.1016/j.scitotenv.2024.173270>.

17. Jakovac, C., Latawiec, A., Lacerda, E., Lucas, I., Korys, K., Iribarrem, A., Malaguti, G., Turner, R., Luisetti, T., & Strassburg, B. (2020). Costs and Carbon Benefits of Mangrove Conservation and Restoration: A Global Analysis. *Ecological Economics*, 176, 106758. <https://doi.org/10.1016/j.ecolecon.2020.106758>.